**Title**

PLC Based Digital Twin, Predictive Fault Detection Sensor and Actuator for Intelligent Industrial Industrial Automation

**Methodology**

1. Sensors are installed on the physical industrial system to continuously measure parameters such as temperature, current, and vibration

2. The sensor signals are conditioned and electrically isolated using suitable circuits, including optocouplers.

3. A PLC collects the sensor data and controls the industrial equipment such as a motor or pump.

4. The PLC transfers real-time operating data to a computer or monitoring platform through a suitable communication protocol.

5. A Digital Twin is developed to represent the real-time condition and behaviour of the physical system.

6. The collected data is analysed to identify abnormal operating conditions and changes in parameter trends.

7. The fault detection module identifies possible faults and provides an early warning before a critical failure occurs.

8. The system displays machine parameters, system health, fault status and predicted faults on a monitoring dashboard.

**Proposed Block Diagram**

Physical System
↓
Sensors
↓
Signal Conditioning
↓
Optocoupler Isolation
↓
PLC
↓
Data Communication

↓
Digital Twin
↓
Data Analysis
↓
Fault Detection & Prediction
↓
Monitoring Dashboard
↓
Warning / Maintenance Alert
↓
PLC Control
↓
Motor / Pump
↓
Physical System

**Proposed Outcome**

The proposed system will provide real-time monitoring of industrial equipment and create a digital representation of the physical system through a Digital Twin. It will detect abnormal operating conditions and predict potential faults using sensor data and operating trends. The system will provide early warnings to the operator, allowing preventive maintenance before major equipment failure.

The expected outcome is a low-cost, modular and scalable predictive maintenance system that can reduce unexpected downtime, improve equipment reliability, increase operational safety and reduce maintenance costs. The system can also be extended to other industrial applications by adding different sensors, equipment models and fault-detection algorithms.